

Bangla Authorship Style Transfer Project Update

Subword Tokenization Experiments

Group Alchemists

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May 4, 2026

Changes Introduced

Subword Tokenization via SentencePiece

- Replaced word-level tokenization with **SentencePiece** (Google's SentencePiece library).
- Applied consistently across both **training** and **inference** pipelines to prevent vocabulary mismatch.

Observed Training Behaviour

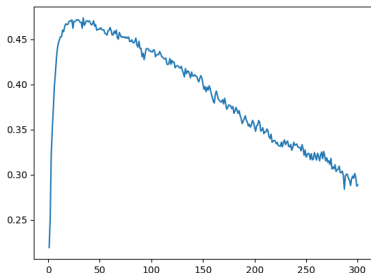
- Discriminator accuracy declined far more steeply than the word-level baseline.
- Epochs increased from **200** $\rightarrow$ **300**, yet accuracy did not recover to the $\sim 20\%$ convergence floor.
- Peak accuracy dropped from **65 %** (word-level) to **47 %** (subword) — the model requires further tuning of optimizer learning rates and λ scheduling to adapt.

Planned Remediation

- Re-tune learning rates and the λ schedule of the gradient-reversal layer to stabilise adversarial training under the new vocabulary.
- We plan to continue working on these improvements after semester examinations.

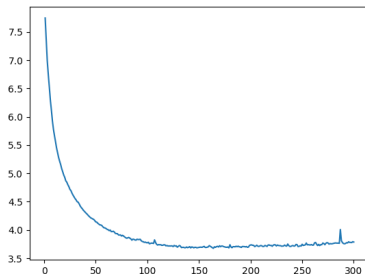
Training Curves

Discriminator Accuracy



Accuracy vs. Epoch

Loss Curve



Total Loss vs. Epoch

- Peak accuracy reached $\approx 47\%$ — well below the word-level baseline of **65%**.
- Accuracy has not converged back to the $\sim 20\%$ floor after 300 epochs, indicating the GRU latent space still retains author-style information.

Model Outputs (Style Transfer)

Tagore

১. কুঁচুলা কানেক জলধি..... কানেক জলধি, কানেক জলধি, কানেক জলধি
২. কানেক জলধি কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি কানেক জলধি

Bibhutibhushan

১. কানেক জলধি কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি
২. কানেক জলধি কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি

Sarat Chandra

১. কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি
২. কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি

Sunil Gangopadhyay

১. কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি
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Satyajit Ray

১. কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি
২. কানেক জলধি..... কানেক জলধি কানেক জলধি কানেক জলধি

The same input sentences used during the original presentation have been retained here for a direct, like-for-like comparison.

Same content → different author styles

❶ **Output repetitions persist.**

Temperature scaling and a repetition penalty were introduced to mitigate this, but the current hyperparameter values appear too relaxed under the subword vocabulary and will need to be re-tuned for the new tokenization scheme.

❷ **Style invariance not yet achieved.**

The discriminator has not converged back to the $\sim 20\%$ floor, meaning the GRU latent representations still carry author-style information. Extending training to ~ 400 epochs is expected to resolve this, but the 5–6 hour retraining time has been deferred for now.

❸ **SentencePiece captures local sub-word dependencies better.**

Qualitative evaluation shows the subword model handles Bangla local word dependencies in a sentence better than word-level tokenization. Sentence-level fluency is still imperfect but appears marginally improved, and we feel should improve further once the issues above are addressed.